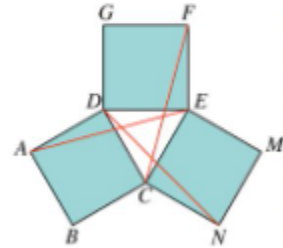


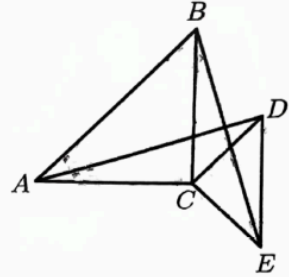
L4

Math Exploration 3

The diagram shows three congruent squares and an equilateral triangle. Prove that $AE = FC = ND$.



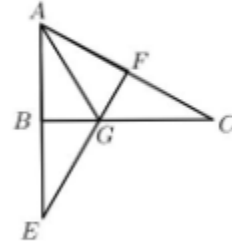
- 9 As shown in the figure below, $\triangle ABC$ and $\triangle ECD$ are isosceles right triangles. $m\angle ACB = m\angle DCE = 90^\circ$, connect $\overline{AD}$ and $\overline{BE}$. Prove that $\triangle ACD \cong \triangle BCE$.



L5

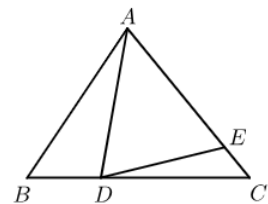
Math Exploration 3

2. In $\triangle ABC$, $m\angle ABC = 90^\circ$. Extend $\overline{AB}$ to E so that $AE = AC$. Create $\overline{EF} \perp \overline{AC}$ at F going through E such that $\overline{EF}$ intersects $\overline{BC}$ at G . Prove that $\overline{AG}$ bisects $\angle BAC$.



Q8

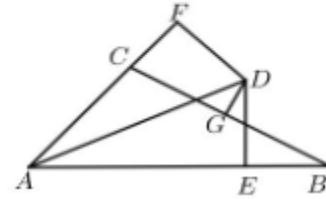
It is given that two base angles of an isosceles triangle are equal to each other. As shown in the diagram, then in $\triangle ABC$, $AB = AC$, $m\angle BAD = 30^\circ$, and $AE = AD$. Then $m\angle EDC = \underline{\hspace{1cm}}$.



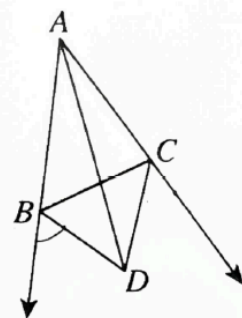
- A) 15
- B) 10
- C) 5
- D) 20

Math Exploration 3

2. As shown in the figure below, $\overline{AD}$ is the angle bisector of $\angle BAC$ and $\overleftrightarrow{DG}$ is perpendicular bisector of $\overline{BC}$. $\overline{AD}$ and $\overleftrightarrow{DG}$ intersect at D . $\overline{DE} \perp \overline{AB}$ at E , $\overline{DF} \perp \overline{AC}$ at F , $AB = 6$, $AC = 3$, then $BE =$ _____. (A little hard, try to connect some points!)

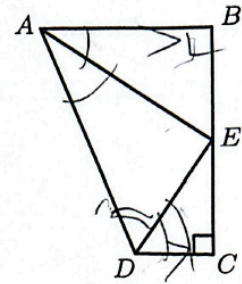


As shown in the figure below, two exterior angle bisectors of $\triangle ABC$ intersect at D .
Prove: $\overline{AD}$ is the angle bisector of $\angle BAC$.



L7

- 3 In the quadrilateral $ABCD$ shown in the figure, $\overline{AB} \parallel \overline{CD}$. If the angle bisector of $\angle DAB$, denoted as $\overline{AE}$, intersects $\overline{BC}$ at E , and $\overline{DE}$ bisects $\angle ADC$. Determine whether the following conclusion is true or false.



1. $CD + AB = BC$
2. E is the midpoint of $\overline{BC}$
3. $\angle AED = 90^\circ$ ✓ cp
4. the area of triangle $ADE = \frac{1}{2}$ the area of quadrilateral $ABCD$

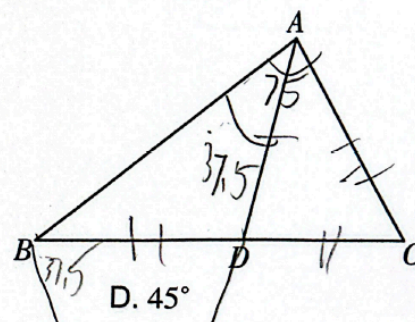
- 4 In triangle ABC , $\overline{AD}$ is the angle bisector of $\angle BAC$, and $AB = AC + CD$. If $m\angle BAC = 75^\circ$, then the measure of $\angle B$ is _____.

A. 25°

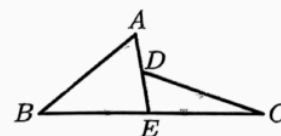
B. 35°

C. 37.5°

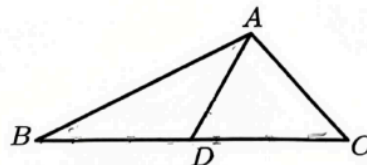
D. 45°



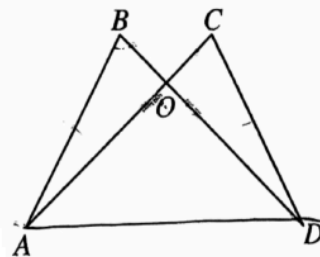
- 5 As shown below, $\angle A = 60^\circ$, point E is a midpoint on line BC . D is a point on line AE . $AB = CD = 3$. Find the value of $AE + DE =$ _____.



- 7 In $\triangle ABC$, $BD = CD$, $\angle DAC = \angle ABC + \angle ACB$. Given that $AD = 3$, find the length of $AB = \underline{\hspace{2cm}}$.



- 10 As shown in the figure below, $AC = DB$, $m\angle B = m\angle C$, prove: $AB = CD$.



L8



Math Exploration 2

2. How many right triangles have integer leg lengths a and b and a hypotenuse of length $b + 1$, where $b < 100$? (2008 AMC 10B Problem, Question #15)

A. 6

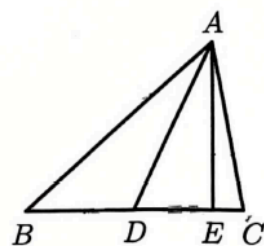
B. 7

C. 8

D. 9

E. 10

- 8 As shown in the figure, in $\triangle ABC$, $\overline{AD}$ is the median from A to $\overline{BC}$, and $\overline{AE} \perp \overline{BC}$ at point E . If $AB = 12$, $BC = 10$, and $AC = 8$, then DE is _____.



Math Exploration 2

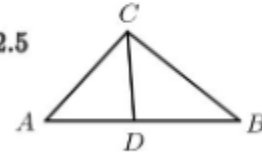
1. In $\triangle ABC$, $\overline{CD}$ is the angle bisector of $\angle ACB$, and $CA = 3$, $CB = 5$, $AB = 6$. The length of $\overline{AD}$ is _____.

A. 2

B. 3

C. $\frac{9}{4}$

D. 2.5

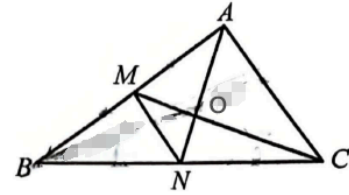


Math Exploration 2

2. Trapezoid $ABCD$ has $\overline{AD} \parallel \overline{BC}$, $BD = 1$, $m\angle DBA = 23^\circ$, and $m\angle BDC = 46^\circ$. The ratio $BC : AD$ is $9 : 5$. What is CD ? (2009 AMC 12B Problem, Question #16)

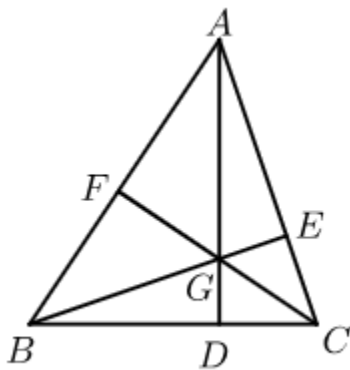
A. $\frac{7}{9}$ B. $\frac{4}{5}$ C. $\frac{13}{15}$ D. $\frac{8}{9}$ E. $\frac{14}{15}$

3. As shown in the figure below, O is the centroid of triangle ABC . $\overline{AN}$ and $\overline{CM}$ intersect at point O . The area of triangle MON is 1. Find the area of triangle ABC .



Q2

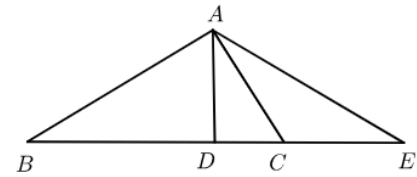
$\overline{AD} \perp \overline{BC}$ at point D , $\overline{BE} \perp \overline{AC}$ at point E , $\overline{CF} \perp \overline{AB}$ at point F , and $\overline{AD}$, $\overline{BE}$, $\overline{CF}$ intersect at point G .



(2) Point A is the orthocenter of triangle ____ .

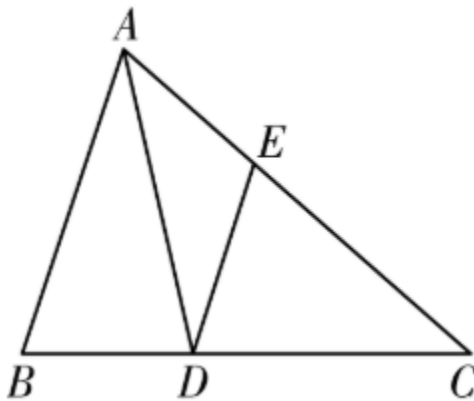
Q4

In the figure, in right triangle ABC , $m\angle BAC = 90^\circ$, $\overline{AD} \perp \overline{BC}$ at D , construct $m\angle EAC = m\angle B$, and the extension of $\overline{AE}$ intersects with $\overline{BC}$ at E . If $AD = 3$ and $DE = 4$, then $DC =$, $AC =$.



Q8

Given the diagram, it is known that $\overline{AD}$ is the angle bisector of $\triangle ABC$, and $\overline{DE} \parallel \overline{AB}$. If $\frac{AE}{EC} = \frac{2}{3}$, find the value of $\frac{AC}{AB}$.



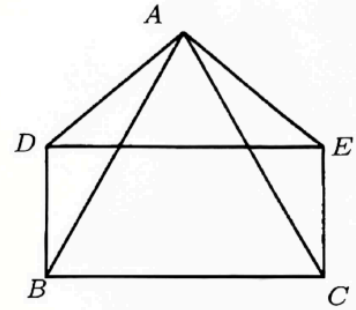
Q10

In triangle ABC , $m\angle A = (x^2 + 6x + 12)^\circ$, $m\angle B = (3x + 8)^\circ$, $m\angle C = (18x)^\circ$, find the longest side of $\triangle ABC$.

L10

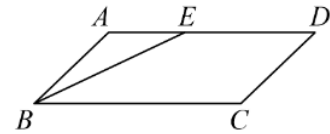
- As shown in the figure, it is given that $AB = AC$, $AD = AE$, $DE = BC$, and $m\angle BAD = m\angle CAE$.

Prove that quadrilateral $BCED$ is a rectangle.



Q2

As shown in the figure, in parallelogram $ABCD$, E is the midpoint of side $\overline{AD}$. If $m\angle ABE = m\angle EBC$, $AB = 2$, then the perimeter of $ABCD$ is _____.



- 10 As shown in the figure below, in parallelogram $ABCD$, extend side $\overline{AB}$ to point E such that $BE = AB$. Draw $\overline{DE}$, $\overline{BD}$, and $\overline{EC}$. $\overline{DE}$ intersects with $\overline{BC}$ at point O . $m\angle BOD = 2m\angle A$. Prove that quadrilateral $BECD$ is a rectangle.

